

Risk■Based Position Sizing — Midas_V2 Guide (v0.3.9 draft)

Purpose. Explain **how** to size trades from a fixed dollar risk and how to extend it later with a conservative “Kelly■lite” boost. No code changes now — this is a reference you can shelf and return to when the core is stable.

1) Why risk■based sizing?

- Keeps losses consistent (e.g., ~\$50 per losing trade) regardless of the ticker’s price.
- Automatically adapts the **share count** to your **stop distance**.
- Easier to budget a **daily max loss** (e.g., \$1,000) and a **per■trade risk** (e.g., \$50).

2) Core definitions

- **Risk per trade (R):** the maximum dollars you’re willing to lose if the stop hits. Example: **R = \$50**.
- **Entry price / Stop price:** planned fill and hard stop.
- **Stop distance per share:** $\text{distance} = |\text{entry} - \text{stop}|$ (add a small buffer for slippage/fees).
- **Shares:** number of shares to buy so that $\text{shares} \times \text{distance} \blacksquare R$.

Base formula (long trades)

$\text{distance_per_share} = (\text{entry} - \text{stop}) + \text{slippage_buffer}$

$\text{shares} = \text{floor}(R / \text{distance_per_share})$

(For shorts, use $(\text{stop} - \text{entry})$ for the distance.)

Example

- Entry **\$5.00**, stop **\$4.88**, buffer **\$0.02** $\Rightarrow$ distance **\$0.14**
- $R = \$50 \Rightarrow \text{shares} = \text{floor}(50 / 0.14) = 357$
- Notional $\approx 357 \times \$5.00 = \$1,785$

3) Guardrails (recommended now)

Add these **caps** to avoid oversizing:

- **max_shares** — e.g., 5,000
- **max_notional** — e.g., \$30,000
- **buying power cap** — shares must fit within $\text{bp_remaining} \times \text{leverage}$
- **daily_max_loss** — stop trading or size down when remaining daily risk $< R$
- **min_size / skip rule** — if computed shares < 1 , skip the trade (stop too tight for chosen R)
- **volatility sanity** — if ATR is huge vs price, optionally scale R down (ATR■based factor)
- **hygiene filters** — price band 1–20, exclude illiquid/warrants/leveraged ETFs, spread/halts checks

Putting it together (conceptual)

...

$\text{shares} = \text{floor}(R / (|\text{entry} - \text{stop}| + \text{buffer}))$

$\text{shares} = \min(\text{shares}, \text{cap_by_max_shares}, \text{cap_by_notional}, \text{cap_by_buying_power})$

if shares < 1: skip

...

4) Daily risk budgeting

If daily_max_loss = \$1,000 and R = \$50: budget $\approx$ 20 full risk trades.

Practical rules:

- Reduce R when drawdown > threshold (e.g., halve R after -\$500).
- Stop trading for the day at $-\$1,000$ realized (or sooner if equity curve deteriorates).

5) “Kellylite” (optional later)

Use a very small Kelly fraction to tilt size when expectancy looks better — but keep it conservative.

Kelly fraction (simplified)

...

$f^* = p - (1 - p) / b$

where p = win rate, b = average win / average loss

...

Use fractional Kelly, e.g., 0.25×Kelly, and clamp to $[0.5, 1.5]$ around base R.

Example

- Suppose $p = 0.60$, $b = 1.0 \Rightarrow f^* = 0.20$ (20% of capital in pure Kelly).
- Use 0.25× $\Rightarrow$ 5% equivalent tilt. In practice, just scale R :

...

$R_{\text{effective}} = \text{clamp}(R_{\text{base}} \times (1 + 0.05), R_{\text{min}}, R_{\text{max}})$

...

Keep $R_{\text{min}}/R_{\text{max}}$ modest (e.g., $\$25 \leq R \leq \75) to avoid big swings.

When to *not* boost

- Fresh month / limited sample size
- After a drawdown day
- When spreads are wide or halts are frequent

6) Integration plan for Midas (later)

No code today. When you're ready:

- **Config keys** (example)

...

risk: {

per_trade_risk: 50,

daily_max_loss: 1000,

```

max_notional: 30000,
max_shares: 5000,
slippage_buffer: 0.02,
kelly_fraction: 0.25, # optional later
kelly_enabled: false,
r_min: 25, r_max: 75
}
...

```

- **Where it plugs in:** the order build step right before entry, using entry/stop and the caps.
- **Logging:** write computed shares, distance, effective R, and which caps limited size.
- **Backtests:** run A/B with sizing="fixed shares" vs "risk based" on the same ranges to verify WR/PnL and drawdowns.

7) Testing checklist

- Unit test the sizing math (edge cases: tiny distance, caps binding, zero/negative values)
- Simulate a day with multiple trades to confirm daily_max_loss halts new entries
- Confirm short trades use proper `distance = stop - entry`
- Verify logs show (entry, stop, distance, shares, cap source)
- Regression run: compare Aug & Sep with risk based sizing **off** first (baseline), then **on**

8) Quick reference (cheat sheet)

- **shares** = `floor( R / (|entry - stop| + buffer) )`
- Apply **max_shares**, **max_notional**, **buying_power** caps
- **Skip** if shares < 1
- **Daily stop** at -\$1,000 (example)
- Optional: **Kelly lite** tilt (small, clamped)

9) Q&A

Q: Can I use this with multiple scenarios?

A: Yes — share a single risk budget and enforce "one position per symbol" across scenarios.

Q: Does this change the signals/entries?

A: No — only the size changes.

Q: Do I need 1sec data first?

A: No — sizing is independent of bar resolution. Keep the minute bar baseline.

Q: When should I enable it?

A: After you finish September regression and tag a stable version.

(Document only. Save under `Docs/versions/6. version v0.3.9 plans/` when convenient.)